

STUDY NOTES — ELABORATED Biological and Anatomical Aspects of Elephants

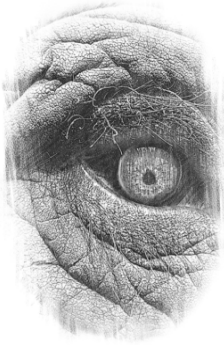
Dr. Parag Nigam, PhD · Wildlife Institute of India, Dehradun

*Training Program: Essentials for Mortality Investigation of Asian Elephant Chhattisgarh Forest Department ·
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Biological and Anatomical Aspects of Elephants

Parag Nigam, PhD
Head, Dept. of Wildlife Health Management
Wildlife Institute of India
nigamp@wii.gov.in

Training Program "Essentials for Mortality Investigation of Asian Elephant," for officials of Chhattisgarh Forest Department, June 5th-6th, 2026, Raigarh, Chhattisgarh



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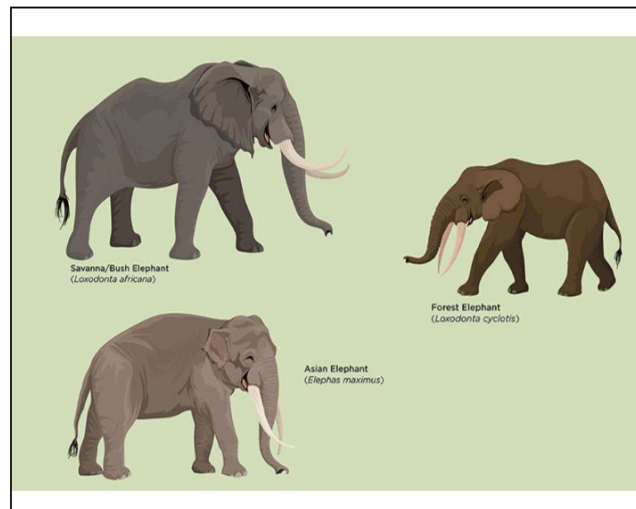
Slide 1 — Title slide



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Slide 2 — Largest terrestrial mammal

1. Taxonomy, Classification & Overview



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Slide 3 — Three living species of elephants

Knowing Elephants



- Family Proboscidae
- Asian elephant with five strains, Indian, Burmese, Ceylonese, Sumatran, Malaysian.
- A bull can stand 9 ft. (2.7 m) in height & weigh between 3-5 tons. Cows can stand 7.5 ft. (2.3 m) in height & weigh between 2.3-4.5 tons.
- Calf at birth weighs 80-100 kgs and has a height of 90-100 cms.
- Poor eye sight, relies on sense of smell and can perceive sound frequencies inaudible to human ear.

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Slide 4 — Key facts about the Asian elephant

1.1 Classification

Rank	Value
Kingdom	Animalia
Phylum	Chordata
Class	Mammalia
Order	Proboscidea (fossil record ~55 million years)
Family	Elephantidae
Genus/Species (Asian)	Elephas maximus Linnaeus, 1758

1.2 Three Living Species

Species	Scientific Name	Key Distinction
Savanna Elephant	<i>Loxodonta africana</i>	Largest; 2 trunk-tip fingers; 4 hindfoot hooves
Forest Elephant	<i>Loxodonta cyclotis</i>	Smaller; rounder ears; rainforest adapted
Asian Elephant	<i>Elephas maximus</i>	1 trunk-tip finger; domed head; 5 hindfoot hooves

1.3 Asian Elephant Subspecies (IUCN 2020)

Subspecies	Range	IUCN Status
<i>E. m. indicus</i> (Indian/Mainland)	South & Southeast Asian mainland	Endangered
<i>E. m. maximus</i> (Sri Lankan)	Sri Lanka	Endangered — largest body size
<i>E. m. sumatranus</i> (Sumatran)	Sumatra, Indonesia	Critically Endangered — smallest
<i>E. m. borneensis</i> (Bornean Pygmy)	Borneo	Endangered — genetically distinct

1.4 Body Size & Senses

Parameter	Bull	Cow
Shoulder height	2.7–3.2 m (9 ft)	2.3–2.5 m (7.5 ft)
Weight	3–5.4 tonnes	2.3–4.2 tonnes
Calf at birth — height/weight	90–100 cm / 80–100 kg	90–100 cm / 80–100 kg

- Olfaction: ~2,000 olfactory receptor genes — most of any mammal; detects water up to 12 km
- Hearing: detects infrasound <20 Hz; communication range up to 10 km
- Touch: seismic sense through feet may detect distant elephant movements

2. The Trunk (Proboscis)



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Slide 5 — The trunk: structure and fine-motor capability

2.1 Structure — Muscular Hydrostat

The trunk is a fusion of the elongated nose and upper lip. It is a MUSCULAR HYDROSTAT — generates all movement through coordinated muscle contractions, with no skeletal support.

- Up to 150,000 separate muscle fascicles (not just 6,000 circular muscles)
- Four muscle groups: longitudinal (elongation), radial (narrowing), transverse (lateral bend), oblique (torsion/rotation)
- No bones, no cartilage — shape changes rely on near-incompressibility of muscle tissue
- A unique proboscis nerve runs along both sides; abundant blood vessels enable engorgement-based stiffening

2.2 Asian vs African Trunk Tip

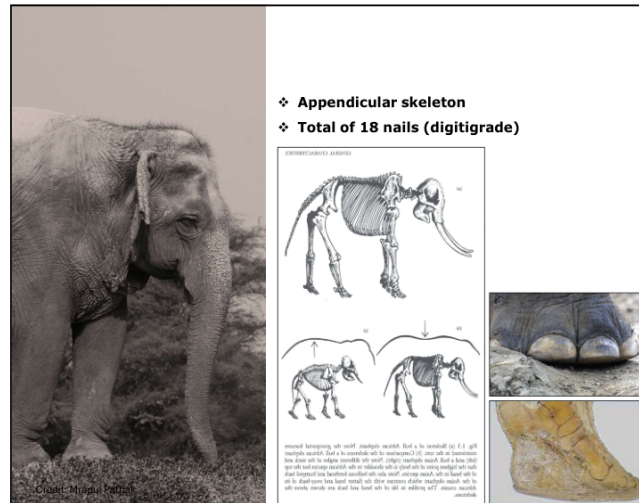
Feature	Asian	African
Finger-like processes	ONE (upper lip only)	TWO (upper + lower lip)
Gripping	Wrap-around grasp	Pinch grip
Hindfoot hooves	5	3

2.3 Functions

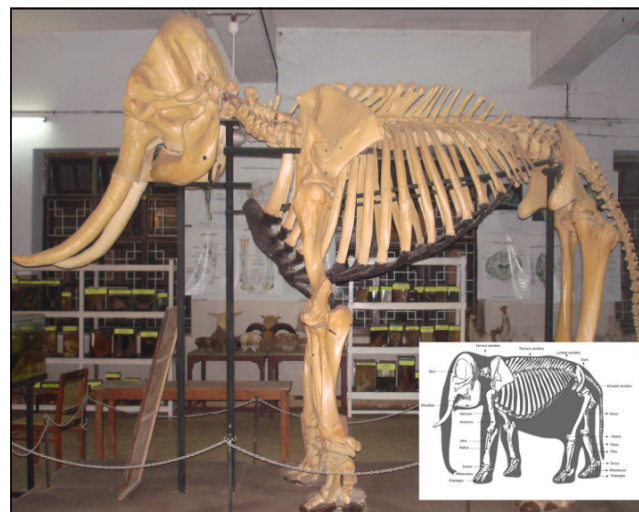
Function	Detail
Respiration	Primary airway — breathing through trunk nostrils
Olfaction	2,000+ olfactory receptor genes; water detected up to 12 km
Water intake	Holds 8–12 litres per fill; for drinking, bathing, spraying
Load carrying	Carries up to 270 kg; picks up objects as small as a pin
Daily feeding	Processes ~150 kg of food daily
Communication	Trumpeting, rumbling, social touch gestures
Combat	Striking, pushing, throwing in male-male conflict

KEY: 150,000 muscle fascicles, no bones — breathes, smells water 12 km away, carries 270 kg, picks up a pin.

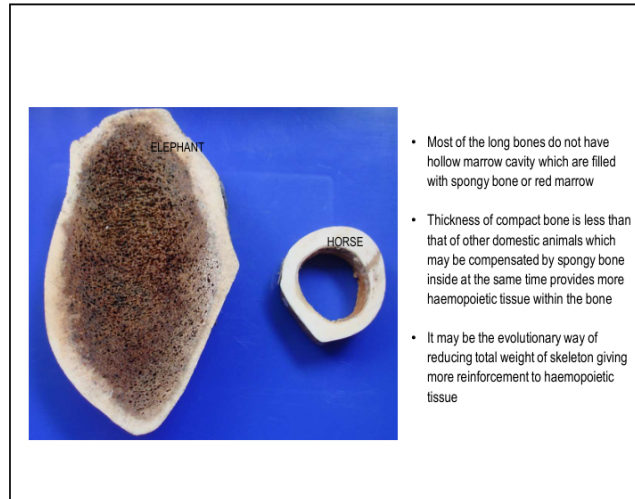
3. Skeletal System



Slide 6 — Appendicular skeleton and 18 nails



Slide 7 — Articulated elephant skeleton



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Slide 8 — Bone cross-section: Elephant vs Horse

3.1 Overview

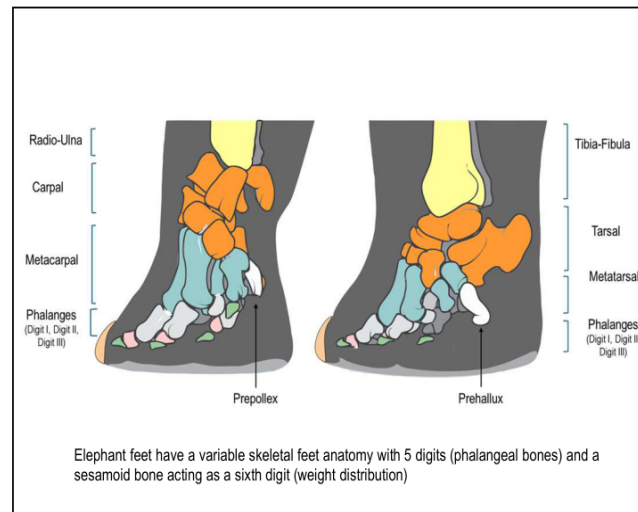
- Total bones: 326–351; Asian elephants: 19–20 rib pairs (African: 21)
- No collarbone — shoulder blade suspended by muscular sling (absorbs impact per step)
- Vertebrae: tight joints limit backbone flexibility; cervical vertebrae elongated for trunk support
- 18 nails total (digitigrade stance); limbs function as vertical pillars

3.2 Bone Microstructure

Feature	Elephant	Horse/Cattle
Medullary cavity	ABSENT — no hollow space	Present — large hollow
Internal fill	Spongy bone + red marrow throughout	Compact cortex, hollow centre
Compact bone	THINNER than domestic animals	Thick cortical bone
Haemopoietic	More abundant throughout bone	Confined to epiphyses

NECROPSY: Thin compact bone + no marrow cavity = NORMAL. Do NOT diagnose as osteoporosis or nutritional bone disease.

4. Foot Anatomy



Slide 9 — Foot skeletal anatomy: Prepollex and Prehallux

4.1 Structure

- Semi-digitigrade posture: walks on toes but appears flat due to fibro-fatty cushion pad
- Forefoot: Radio-Ulna → Carpal → Metacarpal → Phalanges (Digits I–III) + Prepollex
- Hindfoot: Tibia-Fibula → Tarsal → Metatarsal → Phalanges (Digits I–III) + Prehallux

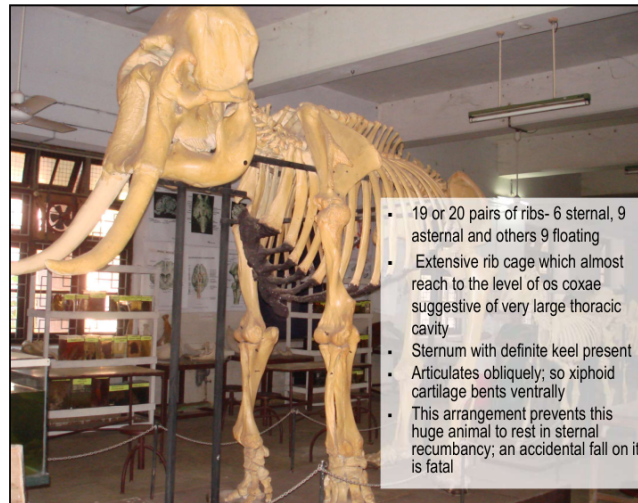
4.2 The Sesamoid "Sixth Digit"

The pre-hallux/pre-pollex is a specialized sesamoid bone embedded in the fat pad — NOT a true articulating toe. It acts as an internal support strut and lever for load distribution.

- Foot pad compresses under load and expands when weight is removed — shock absorption
- Distributes 3–5 tonnes across a wide surface, reducing ground pressure per unit area
- Enables near-silent movement despite enormous body weight
- Front foot circumference (FFC) accounts for 86% of variance in body weight (validated)

FIELD: Front Foot Circumference from ground print → Height = $2 \times \text{FFC}$. Fastest field estimate at mortality site.

5. Thorax & Rib Cage



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Slide 10 — Thoracic skeleton reaching to os coxae level

5.1 Rib Count

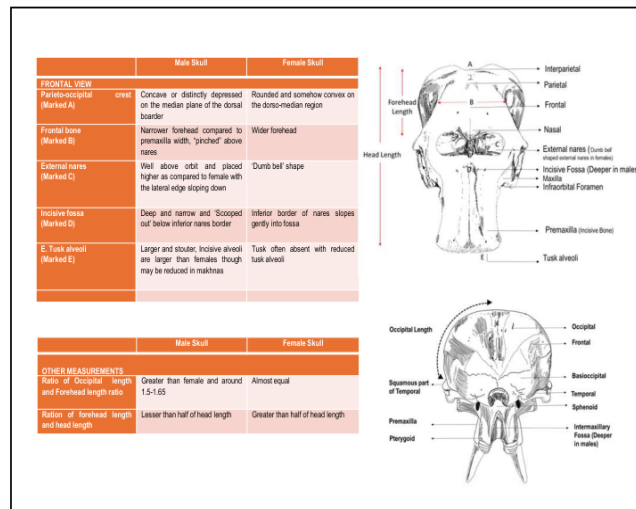
Parameter	Asian Elephant	African Elephant
Total rib pairs	19–20	21
Sternal ribs	6	6
Asternal ribs	9	9
Floating ribs	~4–5	~6
Sternum	Keel present; xiphoid bends ventrally	Similar

5.2 No Pleural Space — Critical Unique Feature

Elephants are the ONLY mammals with no true pleural space. Both pleural layers are fused by dense connective tissue. Lungs are physically adhered to the chest wall. Breathing is driven entirely by diaphragm movement.

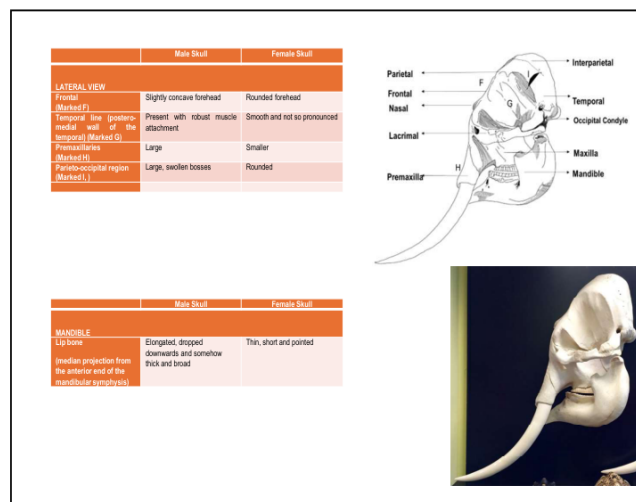
CRITICAL: (1) STERNAL RECUMBENCY = FATAL — abdominal organs crush diaphragm → respiratory arrest. (2) NO THORACOCENTESIS possible. (3) ANAESTHESIA: ventilation must account for diaphragm-driven breathing — IPPV differs from other large mammals. ALWAYS maintain lateral recumbency.

6. Skull Morphology & Sexual Dimorphism



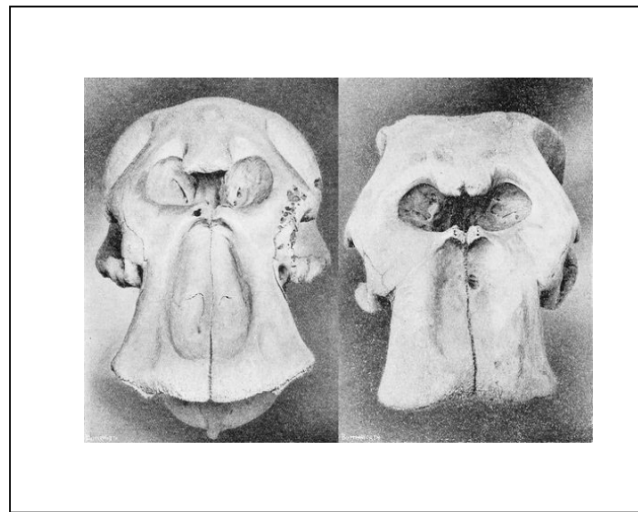
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Slide 11 — Skull: frontal view, male vs female



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Slide 12 — Skull: lateral view and mandible



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Slide 13 — Skull photographs: male (left) vs female (right)

6.1 Skull Architecture

- Skull filled with extensive honeycomb diploic sinuses — reduce weight while maintaining rigidity
- Incisive fossa (premaxilla) = deep depression housing tusk alveoli (roots)
- Apparent forehead "dome" is mostly air sinuses, not brain — true brain case is small and posterior

6.2 Sexual Dimorphism — Frontal View

Feature	Male	Female
Parieto-occipital crest	Concave / depressed on dorsal border	Rounded and convex
Frontal bone	Narrow, "pinched" above nares	Wider forehead
External nares	High above orbit; lateral edge slopes	Dumb-bell shape
Incisive fossa	Deep, narrow, "scooped out"	Slopes gently
Tusk alveoli	Larger and stouter	Absent or reduced

6.3 Lateral View & Mandible

Feature	Male	Female
Frontal	Slightly concave	Rounded
Temporal line	Robust muscle attachment marks	Smooth
Premaxillaries	Large	Smaller
Mandible chin (symphysis)	Elongated, dropped, thick and broad	Thin, short, pointed

PRACTICAL: Skull alone allows sex determination from a decomposed carcass. Dimorphism increases with age — first 2 years skulls are similar between sexes.

7. Age Estimation — Molar Progression

Assessing age



Molar	No. of plates	Appearance age	Replacement age
I	4	4 months	2-2.5 yrs
II	8	6 months	6yrs
III	12	3 yrs	12 yrs
IV	12 (more wide)	8 yrs	25yrs
V	16	20yrs	50-60 yrs
VI	24	40yrs	Life long

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Slide 14 — Molar table and skull cross-section for age assessment

Horizontal molar progression — unique to proboscideans. Six molar sets develop at the back of the jaw and move forward as each preceding molar wears out. Maximum 2 molars in wear simultaneously per half-jaw.

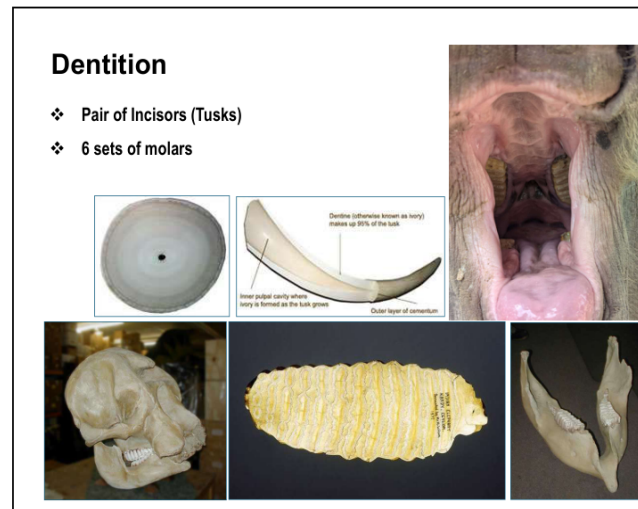
7.1 Molar Table (Asian Elephant)

Molar	Enamel Plates	Appearance Age	Replacement Age	Notes
M1	4	4 months	2–2.5 yrs	Deciduous; worn quickly
M2	8	6 months	6 yrs	Milk tooth
M3	12	3 yrs	12 yrs	First permanent-like
M4	12 (wider)	8 yrs	25 yrs	Wider lamellae
M5	16	20 yrs	50–60 yrs	Longest in use
M6	24	40 yrs	Life long	Never replaced

- African elephant M6 worn by ~47 years; Asian elephant M6 lasts to ~60–70 years
- Count lamellae (enamel plates) + assess wear gradient to estimate age from molar alone

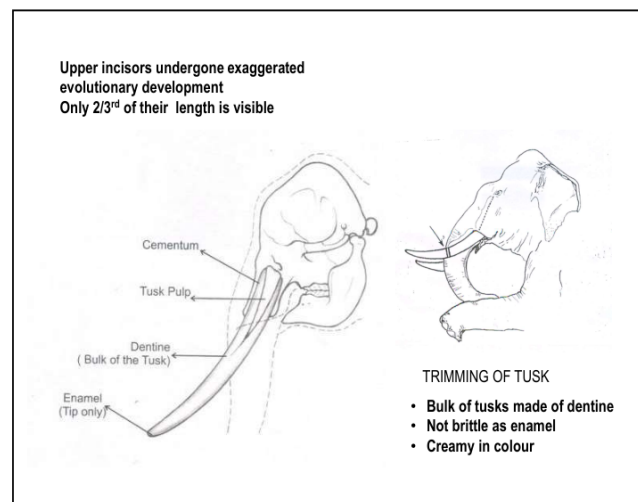
When M6 is exhausted (~60–70 yrs), elephant dies of starvation. Document molar status on ALL mortality forms.

8. Dentition — Tusks & Ivory



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Slide 15 — Dentition: tusks and molar cross-sections



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Slide 16 — Tusk structure: cementum, dentine, pulp, enamel

8.1 Tusk Structure (outer → inner)

Layer	Composition	Notes
Cementum	Bone-like, acellular	Thin outer cover; protects dentine
Enamel	Calcium phosphate crystals	TIP ONLY; wears off quickly
Dentine	Mineralized collagen + hydroxyapatite	~90% of tusk = IVORY
Pulp cavity	Connective tissue, blood vessels, nerves	Where ivory is formed continuously

- Only 2/3 of tusk length is visible; 1/3 is embedded in incisive fossa of premaxilla
- Tusk growth: ~10–15 cm/year in adult bulls; permanent, never shed
- Makhna = tuskless male (common in Sri Lanka; rare in India)

8.2 Schreger Lines



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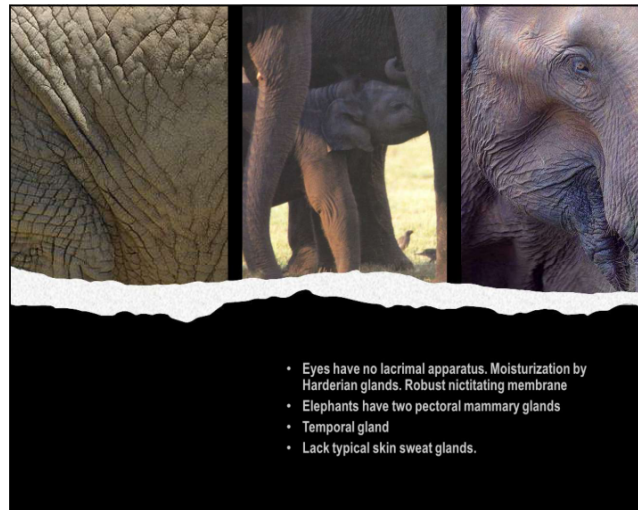
Slide 20 — Schreger lines in ivory cross-section

Schreger lines = characteristic cross-hatching (chevrons/engine-turning) on polished ivory cross-sections. Results from 3D arrangement of mineralized collagen fibrils in dentine.

- Outer Schreger lines: near cementum — clearly visible
- Inner Schreger lines: near pulp cavity — faintly visible
- Elephant ivory: average intersection angle = 115° ("V" chevron shape)
- Mammoth ivory: smaller angle — distinguishable from elephant
- Hippo/walrus/boar ivory: NO Schreger lines — conclusively rules out elephant/mammoth

FORENSIC: Schreger lines confirm elephant ivory identity under UV black light. Gold standard under CITES Appendix I and Wildlife Protection Act 1972 (India). Required in all seizure reports.

9. Skin, Eyes & Integumentary Glands



- Eyes have no lacrimal apparatus. Moisturization by Harderian glands. Robust nictitating membrane
- Elephants have two pectoral mammary glands
- Temporal gland
- Lack typical skin sweat glands.

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Slide 17 — Skin, mammary glands and temporal gland

9.1 Skin

- Deeply wrinkled (up to 2.5 cm thick over back); wrinkles increase surface area for heat loss
- NO body sweat glands — only a few clustered around the toenails
- Thermoregulation: ear flapping (blood cooling), mud/water bathing, shade-seeking

FIELD WARNING: No body sweat glands = high hyperthermia risk during capture/transport. Monitor temperature; spray water on ears; minimize sun exposure.

9.2 Eyes

- No lacrimal apparatus — eye moisturized by Harderian gland in orbit
- Robust nictitating membrane (third eyelid) — moves mediolaterally to clean cornea

9.3 Mammary Glands

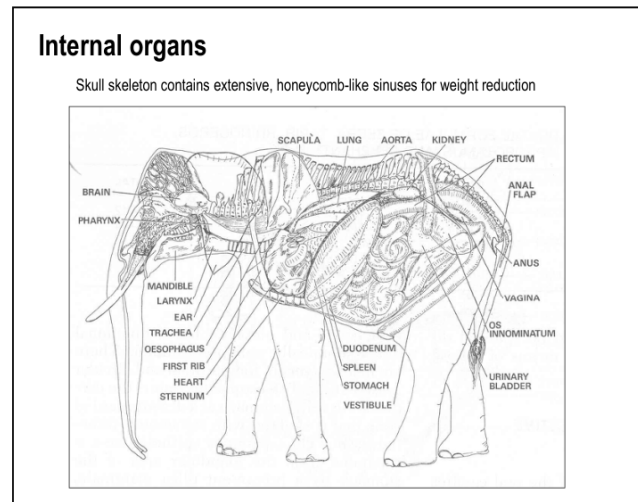
- TWO pectoral mammary glands between front legs — unusual placement (most mammals inguinal/abdominal)
- Calves suckle with mouth (not trunk), reaching up between mother's forelegs
- Lactation: up to 4–5 years; ~11–12 L/day peak milk production

9.4 Temporal Gland — Musth

- Modified APOCRINE SWEAT GLAND between eye and ear; secretes cholesterol, phenol, cresol, farnesol
- In musth bulls: secretes thick "temporin" (dark, tar-like); testosterone up to 60× normal
- Musth signs: temporin secretion + continuous urine dribbling (green) + aggression

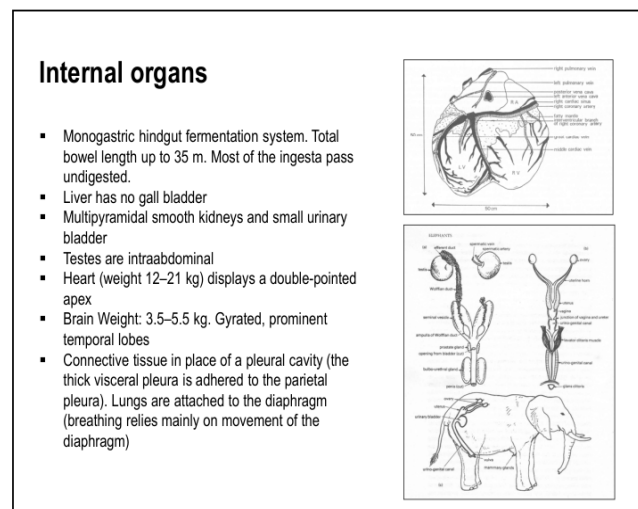
SAFETY: Bull in musth = EXTREMELY DANGEROUS. Dark temporal secretion + urine dribbling = musth. Extreme caution required — even trained captive elephants are unpredictable in musth.

10. Internal Organs



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Slide 18 — Full internal organ diagram (labelled)



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Slide 19 — Heart, reproductive system, respiratory anatomy

10.1 Digestive System

- Monogastric HINDGUT FERMENTATION (like horses — cecum & large intestine ferment cellulose)
- Bowel length: up to 35 m; digestive efficiency ~40%; 150–200 kg plant food daily
- NO GALL BLADDER — bile drains directly from liver into duodenum via bile duct
- Water intake: 80–200 L/day; dung output: ~120 kg/day (16–18 boluses/day)

10.2 Urinary System

- MULTIPYRAMIDAL LOBULATED KIDNEYS (like cetaceans and bears — NOT smooth like most mammals)
- Urine: 10–15 urinations/day; 6–18 L total; green-tinged in musth bulls

10.3 Cardiovascular

Parameter	Value
Heart weight	12–21 kg (up to 30 lbs)
Heart rate	~30 beats/min (resting)
Apex	BIFID (double-pointed) — unique among mammals
Blood volume	~160 L in 5-tonne elephant

- Bifid apex = NORMAL anatomy — do NOT misdiagnose as congenital defect at necropsy

10.4 Reproductive System

Feature	Male	Female
Testes	INTRAABDOMINAL — near kidneys; NO SCROTUM	Two ovaries; cyclical activity
Uterus	—	Bicornuate (two-horned)
Gestation	—	~22 months — longest of any land mammal
Interbirth interval	—	4–5 years

10.5 Brain & Nervous System

Feature	Elephant	Human
Brain weight	3.5–5.5 kg	~1.4 kg
Total neurons	~257 billion	~86 billion
Temporal lobes	Largest of any mammal	Prominent but relatively smaller
Cognitive ability	Tool use, empathy, grief, mirror self-recognition	Yes

10.6 Respiratory

- ONLY mammal with no pleural space — lungs fused to chest wall via connective tissue
- Breathing = diaphragm-driven; very thick diaphragm generates all respiratory pressure

NECROPSY KEY POINTS: No gall bladder | Intraabdominal testes | Bifid heart apex | No pleural cavity | Multipyramidal kidneys | 35 m intestine | Hindgut fermenter | 22-month gestation | 257-billion-neuron brain

11. Field Estimation of Height & Weight

Assessing Height

- ❖ Height = twice circumference of front foot

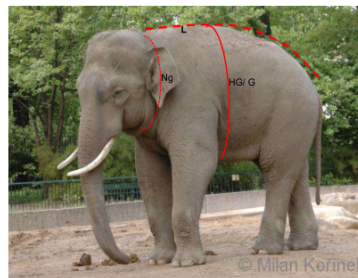


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Slide 21 — Height from front foot circumference

Assessing Weight

- ❖ Weight (Kg) = $12.8 (G + Ng) - 4281$
 - ❖ G = chest girth in cms
 - ❖ Ng = neck girth in cms
- ❖ Weight (Kg) (Male) = $18 (HG) - 3336$
- ❖ Weight (Kg) (Female) = $15 (HG) - 2562$
 - ❖ HG = heart girth in cms
- ❖ Weight (Kg) = $1010 + 0.036 (L \times G)$
 - ❖ L = Body length (cm) from base of forehead to the base of tail
 - ❖ G = Chest girth (cm) measured just caudal to elbow



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Slide 22 — Weight from body girth measurements

11.1 Height Estimation

- Formula: Shoulder Height = $2 \times$ Front Foot Circumference (same units)
- Refined formula: SH (cm) = $1.17 \times$ FFC (cm) + 75.73
- FFC accounts for 86% of variance in body weight (validated in Sumatran elephants)
- FFC vs shoulder height correlation: $R = 0.809$ (strong); FFC vs body length: $R = 0.769$
- Measurable from fresh footprint impression — no direct elephant contact needed

11.2 Weight Formulae

Formula (kg)	Parameters	Application
$12.8 (G + Ng) - 4281$	G = Chest girth (cm), Ng = Neck girth (cm)	General — both sexes

$18 \times \text{HG} - 3336$	HG = Heart girth (cm)	Male-specific
$15 \times \text{HG} - 2562$	HG = Heart girth (cm)	Female-specific
$1010 + 0.036 (\text{L} \times \text{G})$	L = Forehead-to-tail (cm), G = Chest girth (cm)	Two-parameter formula

MANDATORY: Record height (from FFC footprint) and weight (from HG) in ALL mortality investigation forms. Specify which formula used.

12. References & Sources

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Notes from Dr. Parag Nigam PPT (WII) | Web-elaborated with peer-reviewed literature | June 2026 | Raigarh Training Program